Adaptive learning

Mathlete is a web app that teaches math by choosing the next exercise for each learner. It uses a knowledge graph (what you must know first) and a student model (how strong each skill is) to pick the best next action: a short lesson, extra practice, or a quiz. If you answer well, it moves forward. If you struggle, it allows you to pause and revisit the explanation.

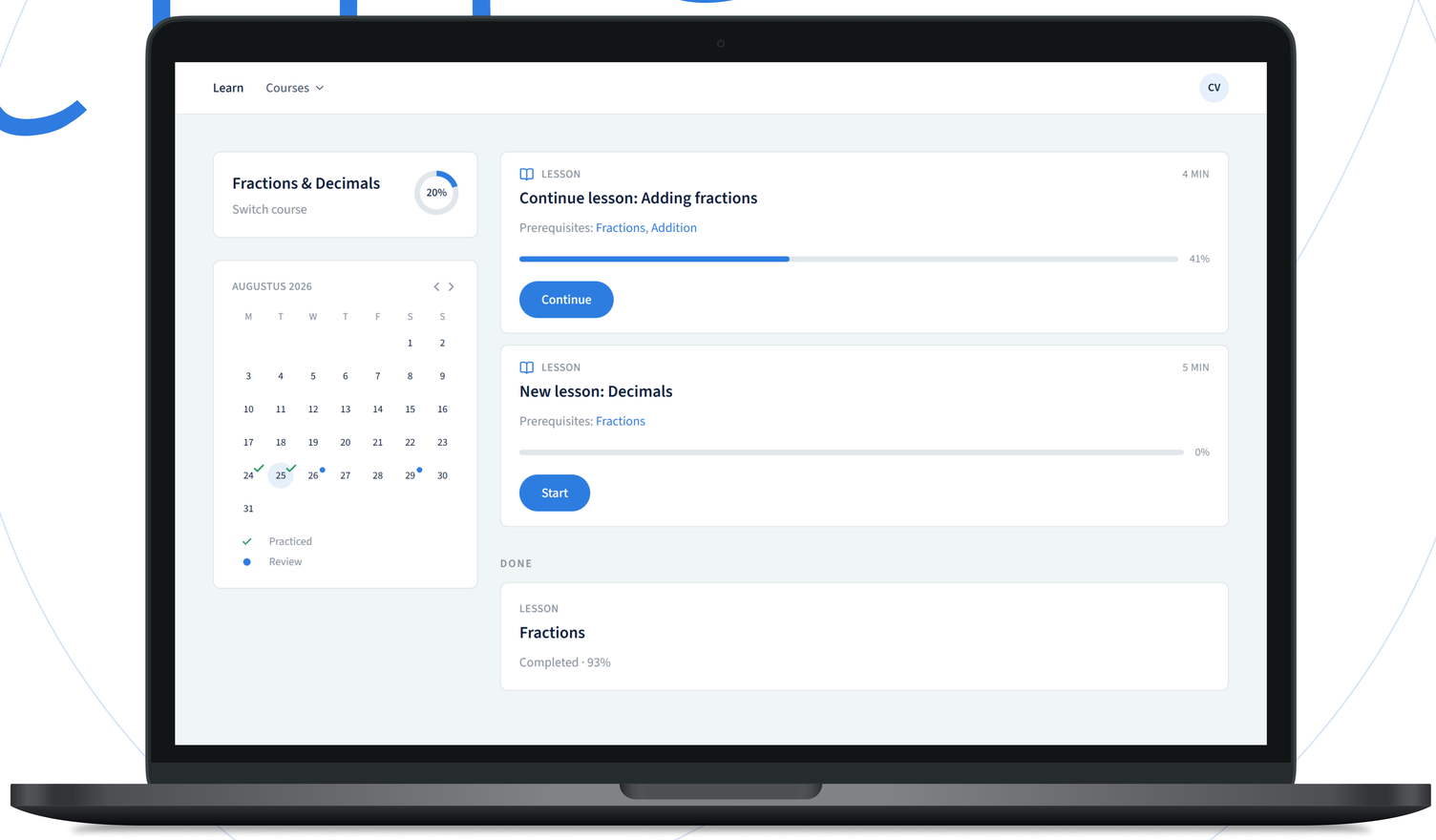
Problem:
The core problem isn't simply “students struggle with math.” It's that traditional learning often follows the same path for everyone, even though students have different strengths, weaknesses, and gaps in knowledge. This can mean students either move on before understanding something or spend time practicing things they already know.

Solution:
Mathlete aims to create a personalized learning path where the learner gets the right type of activity at the right time. Ideally, the result is that students spend more time working on what they actually need, while progressing when they are ready.

Built with:

1. React
2. D3.js
3. FastAPI
4. TypeScript
5. Python
6. Supabase
7. Vite

Mathlete



@Work 4 Graduation Project
Academic Year 2025-2026
Cas Verheye

Artevelde University of Applied Sciences
Associate Degree in Programming